MANGALORE INSTITUTE OF TECHNOLOGY & ENGINEERING
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MINI-PROJECT SYNOPSIS

(25MCSE516)

FakeXpose

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1.Introduction

With the rapid growth of digital media and social networking platforms, the spread of misinformation and fake news has become a major challenge in today's society. News can travel quickly across platforms like WhatsApp, Instagram, and online websites, often without proper verification. This can lead to confusion, panic, and misinformed decisions among the public.

Traditional fake news detection methods rely heavily on manual fact-checking, which is time-consuming and not scalable. Recent approaches use Artificial Intelligence (AI) models to classify news as real or fake, but these systems often depend only on pre-trained data and may not verify information using real-time evidence.

This project focuses on developing an **evidence-based fake news detection system** that combines AI with real-world verification. Instead of relying solely on AI predictions, the system uses AI for understanding and extracting key claims, while validating those claims using trusted news sources, fact-checking APIs, and credibility analysis.

The importance of this project lies in providing a more reliable, transparent, and accurate method for detecting misinformation, helping users make informed decisions and reducing the impact of fake news in society.

2.Objectives

The main objectives of this project are:

- To develop a system that detects fake news using real-world evidence instead of only AI predictions.
- To extract and analyze key claims from user input using AI.
- To verify news using trusted sources and fact-checking APIs.
- To measure similarity between user claims and real news using semantic analysis.
- To implement a credibility scoring mechanism based on source reliability.
- To classify news into Verified, False, or Unverified categories.
- To provide clear explanations for the final decision.
- To design a user-friendly web interface for easy access and usage.

3. Problem Statement

The rapid spread of misinformation through social media, messaging platforms, and online news sources has made it increasingly difficult to distinguish between genuine and fake information. Existing fake news detection systems often rely solely on Artificial Intelligence models, which make predictions based on training data without verifying real-time facts. This can lead to inaccurate or misleading results, as AI models may not reflect current events or validate information against trusted sources.

There is a need for a reliable system that goes beyond AI-based prediction and incorporates real-world evidence, source credibility, and fact-checking mechanisms to verify the authenticity of news. This project aims to address this problem by developing a system that analyzes user-provided information, validates it using trusted sources and APIs, and classifies it as Verified, False, or Unverified, ensuring more accurate and trustworthy results.

4. Existing System

Current fake news detection systems mainly rely on Artificial Intelligence models or manual fact-checking methods. AI-based systems analyze news content and classify it as real or fake based on patterns learned during training. On the other hand, manual fact-checking involves human experts verifying information using trusted sources.

Limitations of Existing System:

- **No real-time verification:** AI models do not always check current or live information
- **Dependence on training data:** Results may be outdated or inaccurate for new events
- **Lack of source validation:** Many systems do not verify credibility of news sources
- **High chance of incorrect predictions:** AI may give confident but wrong answers
- **Slow manual process:** Human fact-checking is time-consuming and not scalable
- **No explanation:** Some systems do not clearly explain why news is classified as fake or real.

These limitations highlight the need for a more reliable system that combines AI with real-world evidence and validation mechanisms.

5. Proposed System

The proposed system is an Evidence-Based Fake News Detection System (FakeXPose) that combines Artificial Intelligence with real-world verification techniques to improve accuracy and reliability. Unlike existing systems that depend only on AI predictions, this system uses AI only for claim extraction and explanation, while the final decision is made using trusted data sources and validation mechanisms.
The system works by taking user input (news or claim), extracting key information using AI, and then verifying it using news APIs, fact-checking services, and semantic similarity analysis.

Advantages over Existing System

- **Real-time verification:** Uses live data from trusted news sources
- **Higher accuracy:** Combines AI with evidence-based validation
- **Source credibility check:** Evaluates reliability of information sources
- **Handles new/unknown news:** Provides "Unverified" instead of wrong results
- **Explainable results:** Gives reasons for classification
- **Reduced AI dependency:** AI is used only as a supporting tool, not the final decision-maker.

6. Scope of the Project

The scope of this project is to develop a web-based fake news detection system that verifies user-provided news or claims using real-time evidence and intelligent validation techniques. The system focuses on analyzing textual input and classifying it as Verified, False, or Unverified based on data from trusted sources.
The project is applicable in areas such as:

- Social media content verification
- Online news validation
- Educational awareness about misinformation
- General public use for checking viral messages

Boundaries of the Project

- The system primarily handles **text-based news/claims**
- Accuracy depends on availability of **real-time data from APIs**
- It does not guarantee 100% correctness but improves reliability using multiple validation layers.

7. Methodology

Approach:

The system uses a hybrid methodology, where:

- AI is used for claim extraction and explanation
- External APIs and algorithms are used for verification and decision-making

Steps Involved:

1. **User Input**
   The user enters news text or a claim through the web interface.
2. **Claim Extraction**
   The AI model processes the input to extract key claims, entities, and important keywords.
3. **Data Retrieval**
   The system fetches related information from:
   - News APIs
   - Fact-checking APIs
   - Trusted online sources
4. **Semantic Analysis**
   Using techniques like Sentence Transformers (SBERT), the system compares the user's claim with retrieved articles to measure similarity.
5. **Credibility Evaluation**
   Sources are analyzed and assigned scores based on their reliability (trusted vs unknown sources).
6. **Rule-Based Decision Engine**
   A scoring system combines:
   - Semantic similarity
   - Source credibility
   - Fact-check result to classify the news as Verified, False, or Unverified.
7. **Explanation Generation**
   The system generates a clear explanation of the result using AI and predefined rules.
8. **Final Output**
   The result along with explanation is displayed to the user.

8. Tools and Technologies Used

Programming Languages:

- **Python** – Backend development and API integration
- **JavaScript** – Frontend interactivity
- **HTML, CSS -** User interface design

Frameworks & Libraries:

- **Flask / FastAPI** – Backend framework for handling requests
- **Sentence Transformers (SBERT)** – Semantic similarity analysis
- **Requests Library** – API communication
- **NumPy / Pandas** – Data processing

APIs Used:

- **OpenAI or Gemini API** – For claim extraction and explanation
- **NewsAPI** – For fetching real-time news articles
- **Google Fact Check Tools API** – For verified fact-check data

Tools / Software:

- **VS Code / PyCharm** – Development environment
- **Git & GitHub** – Version control
- **Web Browser** – Testing and execution

9. System Requirements

9.1 Hardware Requirements:

- **Processor:** Intel i3 / Ryzen 3 or above
- **RAM:** Minimum 4 GB (8 GB recommended)
- **Storage:** At least 20 GB free space
- **Internet:** Stable internet connection (required for API calls)

9.2 Software Requirements

- **Operating System:** Windows / Linux / macOS
- **Programming Language:** Python
- **Frontend:** HTML, CSS, JavaScript
- **Backend Framework:** Flask / FastAPI
- **IDE/Editor:** VS Code / PyCharm
- **APIs Used:**
    - News API
    - Fact Check API
    - AI API (Gemini / ChatGPT)

10 . Module Description

1. User Interface Module

- Allows users to enter news text or claims
- Displays the final result and explanation
- Provides a simple and user-friendly web interface

2. Claim Extraction Module (AI)

- Uses AI to process input text
- Extracts key claims, keywords, and entities

3. Data Retrieval Module

- Fetches related news from **News APIs**
- Retrieves fact-check data from **Fact Check APIs**

4. Semantic Analysis Module

- Uses **Sentence Transformers (SBERT)**
- Compares user claim with retrieved articles

5. Credibility Evaluation Module

- Assigns trust scores to different sources
- Differentiates between trusted and untrusted websites

6. Decision Engine Module

- Combines all scores (similarity, credibility, fact-check results)
- Applies rule-based logic
- Classifies news as:
    - Verified
    - False
    - Unverified

7. Explanation Generator Module

- Generates human-readable explanation
- Explains why the news is classified in a certain category

8. API Integration Module

- Handles communication with external APIs
- Manages API requests and responses efficiently

11. Expected Outcome

The system will:

- Accept user input in the form of news text or claims
- Analyze and extract key information using AI
- Verify the claim using real-time news sources and fact-checking APIs
- Evaluate source credibility and semantic similarity
- Classify the news into:
    - Verified
    - False
    - Unverified
- Generate a clear explanation for the result

Final Result:

The system will help users:

- Make informed decisions before believing or sharing news
- Identify misinformation more accurately
- Understand the reason behind each classification

12.Future Enhancements

- **Image & Video Verification:** Extend the system to detect fake images, deepfakes, and edited videos
- **Multilingual Support:** Support regional languages like Kannada, Hindi, and others
- **Browser Extension:** Develop a plugin to verify news directly while browsing or sharing
- **Social Media Integration:** Analyze trending claims from platforms like Twitter/X and WhatsApp
- **User Feedback System:** Allow users to report or vote on news authenticity
- **Advanced Source Database:** Maintain a dynamic whitelist/blacklist of trusted and fake websites
- **Real-time Alerts:** Notify users about viral fake news or misinformation trends
- **Mobile Application:** Develop an Android/iOS app for wider accessibility

13.References

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